

Lucas Rafael de Andrade

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Education

Getulio Vargas Foundation (FGV)

M.SC. IN APPLIED MATHEMATICS & DATA SCIENCE

- School of Applied Mathematics (EMAp)

Rio de Janeiro, Brazil

2023 – 2024

Getulio Vargas Foundation (FGV)

B.A. IN ECONOMICS

- EPGE Brazilian School of Economics and Finance

Rio de Janeiro, Brazil

2019 – 2022

Federal University of Technology – Paraná (UTFPR)

TECHNICAL DEGREE IN ELECTRONICS

Curitiba, Brazil

2014 – 2018

Experience

Porto Real Asset

DATA ENGINEER | RISK ANALYTICS

São Paulo, Brazil

Feb 2025 – Present

- First data hire at the company; built the entire data infrastructure for risk management in Credit Rights Investment Funds (FIDCs)
- Architected end-to-end ETL pipelines with Apache Airflow: data ingestion via APIs and web scraping, processing of 1.5GB+ datasets, automated report generation and email delivery via Microsoft Graph API
- Developed structured extraction system for 5,000+ Bank Credit Notes (CCBs) PDFs, capturing 30+ fields per document in automated nightly batch processing
- Created aging reports with weighted scoring and stress matrices for liquidity projection, directly used by the investment team for decision-making
- Implemented Machine Learning models to identify default drivers using demographic and geographic variables, with SHAP explainability analysis
- Built financial calculators with market conventions for fixed income asset pricing
- Optimized data processing on a Linux server with only 8GB RAM using profiling, chunking, and memoization techniques

Federal University of Technology – Paraná (UTFPR)

LAB TECHNICIAN INTERN – FOTON LABORATORY

Curitiba, Brazil

Aug 2017 – May 2018

- Processed and analyzed optical fiber data using Bragg Grating technique for high-precision measurements in research environment
- Programmed microcontrollers for millimeter-precision motor control in experimental laboratory equipment

Skills

- Languages:** Python, SQL, R
- Data Engineering:** Apache Airflow, ETL/ELT, Web Scraping, REST APIs, Microsoft Graph API
- Data Analysis:** pandas, NumPy, scikit-learn, statsmodels, pdfplumber, SHAP
- Infrastructure:** Linux, Docker, Git/GitHub, Selenium
- Domain Expertise:** Financial mathematics, fixed income pricing, credit analysis, default modeling

Awards & Publications

- Gold Medal** – Brazilian Physics Olympiad (OBF)
- Master's Thesis:** "Estimation of Spatiotemporal Poisson Processes with Some Missing Location Data" – FGV EMAP, 2024

Languages

Portuguese (Native) • English (Fluent) • Spanish (Intermediate) • Japanese (Intermediate)